

Guanyang Wang

Curriculum Vitae

110 Frelinghuysen Road

Piscataway, NJ 08854

☎ (650) 422 8730

✉ guanyang1994@gmail.com

🌐 guanyangwang.github.io/

Positions

- 2026–Present **Associate Professor**, *Department of Statistics, Rutgers University*
Graduate Faculty: Department of Computer Science, Rutgers University
Elected Member: Command, Control, and Interoperability Center for Advanced Data Analysis
Member: Center for Discrete Mathematics and Theoretical Computer Science (DIMACS)
- 2020–2026 **Assistant Professor**, *Department of Statistics, Rutgers University*

Research Interests

Generative AI (diffusion models and world models), tabular foundation models, AI4Math, Monte Carlo methods, and quantum computing

Education

- 2015–2020 **Ph.D.**, *Department of Mathematics, Stanford University*
Advisor: Professor Persi Diaconis
Ph.D. minor in Statistics
Thesis: *Topics in Markov chain Monte Carlo Methods, with applications in statistics*
- 2011–2015 **B.S.**, *School of Mathematical Sciences, University of Science and Technology of China*
Summa cum laude
Guo Moruo Scholarship (highest honor for USTC students)

Awards and Honors

- 2026 Presidential Fellowship for Teaching Excellence, *Rutgers University*
- 2023 Blackwell-Rosenbluth Award, *Junior Section of the International Society for Bayesian Analysis (j-ISBA)*; one of six recipients
- 2022 Adobe Data Science Research Award, *Adobe Research*; with Peng Zhang; one of four recipients
- 2022 Ralph E. Powe Junior Faculty Enhancement Award, *Oak Ridge Associated Universities*; one of 37 nationwide awards across multiple disciplines and one of two awarded statisticians
- 2022 IMS New Researcher Travel Award, *Institute of Mathematical Statistics (IMS)*
- 2016 Halsey L. Royden, Jr. Fellowship, *Stanford University*
- 2014 Guo Moruo Scholarship (top 1%), *University of Science and Technology of China*
- 2013 National Scholarship (top 1%), *University of Science and Technology of China*

Grants Awarded

- NSF **Collaborative Research: FET: Medium: Quantum Monte Carlo Speed Ups for Multilevel Computations and Other Statistical Algorithms**
National Science Foundation, CCF-2403007, 2024–2027
Role: **Co-PI**; PI: Mario Szegedy; other Co-PI: Jose Blanchet
Total: \$1,198,712; my share: **\$400,000**
- NSF **Scalable Algorithm Design for Unbiased Estimation via Couplings of Markov Chain Monte Carlo Methods**
National Science Foundation, DMS-2210849, 2022–2025
Role: **PI**
Total: **\$200,000**
- Adobe **Adobe Data Science Research Award**
Adobe Research; with Peng Zhang
Role: **PI**
Total: **\$50,000**
- ORAU **Ralph E. Powe Junior Faculty Enhancement Award**
Oak Ridge Associated Universities, 2022
Role: **PI**
Total: **\$10,000**
- Rutgers **Provost's COVID Impact Faculty Grant**
Rutgers University, 2021
Role: **PI**
Total: **\$5,000**

Published and Accepted Papers

★ = alphabetical author order; † = equal contribution;
◇ = corresponding author; ★★ = student first author

- SODA **Spectral Gap for the Binary Fixed-Margin Swap Chain**
2027 Weibo Fu, Qian Qin, Guanyang Wang[★].
Annual ACM-SIAM Symposium on Discrete Algorithms (SODA), 2027.
Accepted.
- NeurIPS **TabClustPFN: A Prior-Fitted Network for Tabular Data Clustering**
2026 Tianqi Zhao, Guanyang Wang, Yan Shuo Tan, Qiong Zhang.
Neural Information Processing Systems (NeurIPS), 2026.
- AoS **On importance sampling and independent Metropolis-Hastings with an unbounded weight function**
Accepted George Deligiannidis, Pierre E. Jacob, El Mahdi Khribch, Guanyang Wang[★].
Annals of Statistics.
Accepted.
- ICML **Weak Diffusion Priors Can Still Achieve Strong Inverse-Problem Performance**
2026 Jing Jia[†], Wei Yuan[†], Sifan Liu, Liyue Shen, Guanyang Wang[◇].
International Conference on Machine Learning (ICML), 2026.
Spotlight; top 2.2%.

- ICML 2026 **Markov chain Monte Carlo without evaluating the target: an auxiliary variable approach**
Wei Yuan^{**}, Guanyang Wang[◇].
International Conference on Machine Learning (ICML), 2026.
Oral presentation; top 0.7%.
- ICML 2026 **Optimal Quantum Speedups for Repeatedly Nested Expectation Estimation**
Yihang Sun, Guanyang Wang, Jose Blanchet.
International Conference on Machine Learning (ICML), 2026.
- ICLR 2026 **Antithetic Noise in Diffusion Models**
Jing Jia^{**}, Sifan Liu[◇], Bowen Song, Wei Yuan, Liyue Shen[◇], Guanyang Wang[◇].
International Conference on Learning Representations (ICLR), 2026.
- AoAP 2026 **A phase transition in sampling from Restricted Boltzmann Machines**
Youngwoo Kwon, Qian Qin, Guanyang Wang^{*}, Yuchen Wei.
Annals of Applied Probability 36(2): 1615–1652, 2026.
- NeurIPS 2025 **Quantum speedup of non-linear Monte Carlo problems**
Jose Blanchet, Yassine Hamoudi, Mario Szegedy, Guanyang Wang^{*}.
Neural Information Processing Systems (NeurIPS), 2025.
Spotlight; top 3.2%.
- NeurIPS 2025 **CCS: Controllable and Constrained Sampling with Diffusion Models via Initial Noise Perturbation**
Bowen Song, Zecheng Zhang, Zhaoxu Luo, Jason Hu, Wei Yuan, Jing Jia, Zhengxu Tang, Guanyang Wang[◇], Liyue Shen[◇].
Neural Information Processing Systems (NeurIPS), 2025.
- AoS 2025 **Spectral gap bounds for reversible hybrid Gibbs chains**
Qian Qin, Nianqiao Ju, Guanyang Wang.
Annals of Statistics 53(4): 1613–1638, 2025.
- WSC 2025 **Connecting Quantum Computing with Classical Stochastic Simulation**
Jose Blanchet, Mark S. Squillante, Mario Szegedy, Guanyang Wang^{*}.
Winter Simulation Conference (WSC), 2025.
- AISTATS 2025 **Differentially Private Range Queries with Correlated Input Perturbation**
Prathamesh Dharangutte, Ruobin Gong, Jie Gao, Guanyang Wang^{*}.
International Conference on Artificial Intelligence and Statistics (AISTATS), 2025.
- AoAP 2024 **Repeated Averages on Graphs**
Ramis Movassagh, Mario Szegedy, Guanyang Wang^{*}.
Annals of Applied Probability 34(4): 3781–3819, 2024.
- AoAP 2024 **Spectral Telescope: Convergence Rate Bounds for Random-Scan Gibbs Samplers Based on a Hierarchical Structure**
Qian Qin, Guanyang Wang^{*}.
Annals of Applied Probability 34(1B): 1319–1349, 2024.
- AIHP 2024 **Metropolis-Hastings transition kernel couplings**
John O’Leary, Guanyang Wang^{*}.
Annales de l’Institut Henri Poincaré (B), Probabilités et Statistiques 60(2): 1101–1124, 2024.

- PLOS CS 2024 **Pattern detection in bipartite networks: a review of terminology, applications, and methods**
With Zachary Neal, Giovanni Strona, and 8 others.
PLOS Complex Systems 1(2): e0000010, 2024.
- JMLR 2023 **Unbiased Multilevel Monte Carlo estimators for intractable distributions: MLMC meets MCMC**
Tianze Wang^{**}, Guanyang Wang[◇].
Journal of Machine Learning Research 24(249): 1–40, 2023.
- ICML 2023 **Optimal randomized multilevel Monte Carlo for repeatedly nested expectations**
Yasa Syed^{**}, Guanyang Wang[◇].
International Conference on Machine Learning (ICML), PMLR 202: 33343–33364, 2023.
Short version; journal version to be submitted.
- PNAS 2023 **Prediction of Enzyme Specificity using Protein Graph Convolutional Neural Networks**
Changpeng Lu, Joseph H. Lubin, Samuel Z. Stentz, Guanyang Wang, Sijian Wang, Sagar D. Khare.
Proceedings of the National Academy of Sciences 120(39): e2303590120, 2023.
- SPA 2023 **Unbiased Optimal Stopping via the MUSE**
Zhengqing Zhou[†], Guanyang Wang[†], Jose Blanchet, Peter Glynn.
Stochastic Processes and their Applications 166: 104088, 2023.
- RS&A 2023 **On the Minimax Spherical Designs**
Weibo Fu, Guanyang Wang^{*}, Jun Yan.
Random Structures & Algorithms 62(1): 131–154, 2023.
- Bernoulli 2022 **Exact convergence rate analysis of the Independent Metropolis-Hastings algorithms**
Guanyang Wang.
Bernoulli 28(3): 2012–2033, 2022.
- Bernoulli 2022 **On the theoretical properties of the exchange algorithm**
Guanyang Wang.
Bernoulli 28(3): 1935–1960, 2022.
- JASA 2021 **Discussion of ‘A Gibbs sampler for a class of random convex polytopes’**
Persi Diaconis, Guanyang Wang^{*}.
Journal of the American Statistical Association 116(535): 1193–1195, 2021.
Invited discussion.
- Nat. Commun. 2021 **Integrated cooling (i-Cool) textile of heat conduction and sweat transportation for personal perspiration management**
With Yucan Peng and 16 others.
Nature Communications 12(1): 6122, 2021.
- AISTATS 2021 **Maximal couplings of the Metropolis-Hastings algorithm**
Guanyang Wang[†], John O’Leary[†], Pierre Jacob.
International Conference on Artificial Intelligence and Statistics (AISTATS), PMLR 130: 1225–1233, 2021.
Oral presentation; top 3% of submissions.

- EJS **A Fast MCMC algorithm for the uniform sampling of binary matrices with fixed margins**
 2020 Guanyang Wang.
Electronic Journal of Statistics 14(1): 1690–1706, 2020.
- ECP **Expectation of the largest betting size in Labouchere system**
 2019 Yanjun Han, Guanyang Wang*.
Electronic Communications in Probability 24: 1–10, 2019.
- AMSA **Bayesian Goodness of Fit Test: A Conversation for David Mumford**
 2018 Persi Diaconis, Guanyang Wang*.
Annals of Mathematical Sciences and Applications 3(1): 287–308, 2018.

Preprints

- 2026 **Architecture Alignment With Sparse Priors in Tabular Foundation Models**
 Tianqi Zhao, Tianyi Zhuang, Shuo Duan, Guanyang Wang, Yan Shuo Tan, Qiong Zhang.
arXiv:2609.36883.
- 2026 **Unlocking Few-Step Diffusion for Faithful Previews**
 Jing Jia, Sifan Liu, Guanyang Wang.
arXiv:2609.34406.
- 2026 **Sharp small-deviation inequalities for sums of independent nonnegative random variables**
 Weibo Fu, Yanjun Han, Guanyang Wang, Jun Yan, Peng Zhang, Zhengqing Zhou.
arXiv:2607.23980.
- 2026 **Couple to Control: Joint Initial Noise Design in Diffusion Models**
 Jing Jia, Liyue Shen, Guanyang Wang.
arXiv:2605.11311.
- 2024 **Putting all eggs in one basket: some insights from a correlation inequality**
 Pradeep Dubey, Siddhartha Sahi, Guanyang Wang*.
arXiv:2403.15957.
 Under review.
- 2024 **Quantum Monte Carlo with quadratic speedup under infinite variance**
 Jose Blanchet, Mario Szegedy, Guanyang Wang*.
arXiv:2401.07497.
- 2024 **When are Unbiased Monte Carlo Estimators More Preferable than Biased Ones?**
 Guanyang Wang, Jose Blanchet, Peter Glynn.
arXiv:2404.01431.

Mentoring

Ph.D. Advising

- 2021–2025 **Yasa Syed**: Ph.D. in Statistics, Rutgers University
Thesis: *Bias and Risk in Modern Stochastic Algorithms: From Multilevel Monte Carlo to Robust Accelerated Optimization*
First position: Quantitative Researcher, Venerable
- 2022–2026 **Wei Yuan**: Ph.D. in Statistics, Rutgers University (2026)
Thesis: *Bayesian Inference Under Intractable Models and Imperfect Priors*
First position: Senior Computer Vision Engineer, Honor Device Co., Ltd.
- 2023–Present **Budhaditya Halder**: Fifth-year Ph.D. student in Statistics, Rutgers University
- 2025–Present **Jing Jia**: Third-year Ph.D. student in Computer Science, Rutgers University

Research Project Supervision

- 2022–2025 **Yuchen Wei**: Ph.D. in Mathematics, Rutgers University
- 2021–2023 **Tianze Wang**: Ph.D. student in Statistics, Rutgers University

Undergraduate and Master's Advising

- 2022–2023 **Qiru Pan**: Master's student in Statistics, Rutgers University
- 2022 **Yingshi Chen**: Undergraduate student in Statistics, Rutgers University

Ph.D. Thesis Committees

Zexi Song (Statistics), **Ryumei Nakada** (Statistics), **Yuze Zhou** (Statistics)

Ph.D. Qualifying Examination Committees

Zhe Zhang (Statistics), **Yuze Zhou** (Statistics), **Daniel Shidi Wu** (Plant Biology),
Ryumei Nakada (Statistics), **Xiangrui Kong** (Statistics)

Invited Talks

- 2026.12 Department Colloquium, Lehigh University
- 2026.11 ARC Colloquium, Georgia Tech
- 2026.10 Department Seminar, Department of Statistics, University of Illinois Urbana-Champaign
- 2026.07 EAC-ISBA Conference, Kunming, China
- 2026.07 ICML Oral Presentation, Seoul, Korea
- 2026.06 ICSA Applied Statistics Symposium, George Mason University, Arlington, Virginia
- 2026.05 New England Statistics Symposium, University of Connecticut, Storrs, CT
- 2026.03 Signal and Information Processing (SIP) Seminar, Rutgers University
- 2026.03 INFORMS Optimization Society Conference (IOS), Atlanta
- 2025.12 Conference "Winter Simulation Conference", Seattle
- 2025.12 Conference "CMStatistics", virtual talk
- 2025.11 Department Seminar, Department of Computer Science, Rutgers University
- 2025.10 Conference "SIAM New York-New Jersey-Pennsylvania Section Conference", Pennsylvania State University, Pennsylvania

2025.09 Conference "Fast and Curious II: MCMC in Action", University of Toronto, Toronto, Canada

2025.08 Conference "8th International Conference on Econometrics and Statistics", virtual talk

2025.07 Joint Conference on Statistics and Data Science, Hangzhou, China

2025.07 INFORMS Applied Probability Society Conference, Georgia Tech, Atlanta

2025.06 Bayes Comp, Singapore

2025.06 Institute of Industrial and Systems Engineers (IISE) Annual Conference, Atlanta

2025.05 Department Seminar, Department of Quantitative Theory & Methods, Emory University

2025.05 Reading group, Flatiron Institute, New York

2025.01 Workshop "Computational Methods in Bayesian Statistics", University of Florida

2024.12 International Conference on Statistics and Data Science (ICSIDS), Nice, France

2024.11 KU Probability and Statistics Conference on Stochastic Analysis and Related Areas, University of Kansas

2024.08 Joint Statistical Meeting (JSM), Portland

2024.07 Conference "7th International Conference on Econometrics and Statistics", virtual talk

2024.06 Conference "EAC-ISBA", virtual talk

2024.04 Mostly Monte Carlo and All About That Bayes joint seminar, PariSanté Campus, France

2024.04 Tutorial Talk, Diffusion Model Reading Group, Department of Electrical and Computer Engineering, University of Michigan

2024.02 Department Seminar, Department of Statistics, Rutgers University

2024.02 Econometrics and Statistics Colloquium, The University of Chicago Booth School of Business

2024.02 Department Seminar, Department of Statistical Science, Duke University

2024.01 Department Seminar, Department of Statistics, Rice University

2024.01 Department Seminar, Department of Statistics, University of Michigan

2023.12 Conference "IMS International Conference on Statistics and Data Science (ICSIDS)", Lisbon

2023.11 Conference "The Bayesian Young Statisticians Meeting (BAYSM 2023)", virtual talk

2023.10 Department Seminar, School of Management Sciences and Information Systems, Rutgers Business School

2023.07 Conference "The 9th International Forum on Statistics", Beijing

2023.06 New England Statistics Symposium (NESS), virtual talk

2023.05 Conference "The Fast and the Curious: Modern Markov chain Monte Carlo", University of Minnesota

2023.05 Discussant, Conference "Recent Advances in Statistics and Data Science", Rutgers University

2023.03 Department Seminar, Department of Statistics, University of Iowa

2022.11 Department Seminar, Department of Statistics, Colorado State University

2022.09 Hon Hai (Foxconn) Quantum Computing Research Center, Taipei

2022.05 The Alan Turing Institute, U.K.

2022.05 New England Statistics Symposium (NESS)

- 2022.04 Department Seminar, Department of Statistics, Texas A&M University
- 2022.04 Discussant, Conference on Advances in Bayesian and Frequentist Statistics: Celebration of the 80th Birthday of Professor William E. Strawderman, Rutgers University
- 2022.03 Department Seminar, Department of Statistics, the Wharton School, University of Pennsylvania
- 2022.02 Department Seminar, Department of Mathematics, Rutgers University-Camden Campus
- 2021.10 Department Seminar, Department of Statistics, University of Minnesota
- 2021.08 Discussant on paper 'A Gibbs sampler for a class of random convex polytopes', Joint Statistical Meeting
- 2021.08 Session: Advances in MCMC Theory and Practice, Joint Statistical Meeting
- 2021.04 The 23rd International Conference on Artificial Intelligence and Statistics 2021
- 2020.07 Mozi Forum, Department of Computer Science, University of Science and Technology of China
- 2020.01 Department Seminar, Department of Statistics, Rutgers University
- 2020.01 Department Seminar, Department of Statistics, University of Florida
- 2019.05 Guest Lecture for Modern Markov Chains, Stanford University
- 2018.09 Probability Research Seminar, Department of Mathematics, University of Science and Technology of China

Professional Activities

Journal Editorial Board

- 2026–Present Associate Editor, Mathematics of Operations Research

Area Chair

- 2026 NeurIPS

Organizer

- 2026 Co-organizer of Quantum Metrics and Performance Evaluation Workshop, a SIGMETRICS 2026 workshop
- 2024–Present Co-organizer of the International Seminar on Monte Carlo Methods, a weekly online seminar on Monte Carlo methods
- 2021 JSM invited session "Inference, optimization, and computation on discrete structures"

Session Chair

- 2023 EAC-ISBA invited session "Recent Developments in Markov Chain Monte Carlo"
- 2023 Conference "The Fast and the Curious: Modern Markov chain Monte Carlo"
- 2021 JSM invited session "Inference, optimization, and computation on discrete structures"

External Committee Service

- 2024 American Statistical Association student paper competition (Section on Bayesian Statistical Science)
- 2024 Scientific committee for the Blackwell-Rosenbluth Award (International Society for Bayesian Analysis)

- 2023–2024 Best Student Paper Competition selection committee (The Applied Probability Society of Institute for Operations Research and the Management Sciences (INFORMS))
- 2023 Student Poster Competition committee (Conference “Recent Advances in Statistics and Data Science”, Piscataway, New Jersey, U.S.)
- 2022 Student Poster Selection committee (New England Statistics Symposium, Storrs, Connecticut, U.S.)

Department and University Service at Rutgers

- 2025–Present Seminar Chair, Department of Statistics
- 2025–Present Ph.D. Examination Committee, Department of Statistics
- 2025–Present Data Science Advisors Committee, Department of Statistics
- 2023–Present Graduate Curriculum Committee, Department of Statistics
- 2023–2025 Department Website Committee, Department of Statistics
- 2023–2024 Undergraduate Studies Committee, Department of Statistics
- 2020–2024 Ph.D. Admissions Committee, Department of Statistics
- 2020–2023 Ph.D. Examination Committee, Department of Statistics
- 2020–2023 Master’s Examination Committee, Department of Statistics
- 2020–2023 Social and Retreat Committee, Department of Statistics
- 2022 Internal Reviewer for Powe Jr. Faculty Enhancement Awards (2023), Rutgers University
- 2021–2022 Conference Organization Committee, Department of Statistics

Peer Review

- Journals Referee for Annals of Applied Probability, Journal of the Royal Statistical Society: Series B (4), Journal of the American Statistical Association (3), Journal of Machine Learning Research, Nature Communications, the Bernoulli Journal (2), Mathematics of Operations Research (2), SIAM Journal on Scientific Computing, SIAM Journal on Uncertainty Quantification, Journal of Computational Physics, Random Structures and Algorithms, Electronic Journal of Probability (3), Electronic Journal of Statistics, Statistics and Computing (4), Journal of Computational and Graphical Statistics (3), Stochastic Processes and their Applications, Proceedings of the Royal Society A, Statistica Sinica, Advances in Applied Probability (2), Statistics and Probability Letters, Environmental and Ecological Statistics
- Conferences NeurIPS 2025 (5), COLT 2025 (2), UAI 2025, STOC 2024 (2), Theory of Quantum Computation (TQC) 2024, AISTATS 2021 (4), Pacific Symposium on Biocomputing 2022
- Other Reviewer for MathSciNet

Teaching Experience

Rutgers University (Instructor)

- Stat 592 Theory of Probability (Fall 2025)
- Stat 486 Applied Statistical Learning (Fall 2023, Fall 2024)
- Stat 681 Advanced Probability Theory II (Spring 2023)
- Stat 680 Advanced Probability Theory I (Fall 2021, Fall 2022)
- Stat 654 Stochastic Processes (Spring 2021, Spring 2022)
- Stat 582 Introduction to Methods and Theory of Probability (Fall 2020)

Stanford University (Teaching Assistant)

- Math 21 Calculus (multiple offerings)
- Math 51 Linear Algebra and Multivariate Calculus (multiple offerings)
- Math 52 Multivariate Integral and Calculus (Spring 2016)
- Math 104 Applied Matrix Theory (Spring 2019)
- Math 113 Linear Algebra and Matrix Theory (multiple offerings)
- Math 131P Partial Differential Equations (Winter 2017)

University of Science and Technology of China (Teaching Assistant)

- Probability Theory and Applications (Spring 2014)